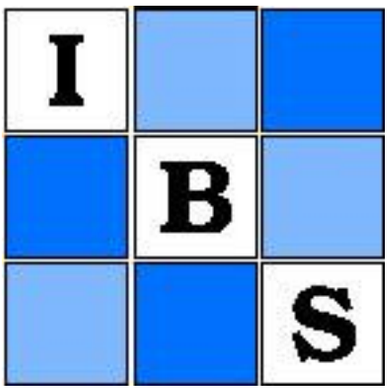


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Author(s): David W. Goodall

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A NEW SIMILARITY INDEX BASED ON PROBABILITY

DAVID W. GOODALL

C. S. I. R. O. Division of Mathematical Statistics, Western Australian Regional Laboratory, Nedlands, Western Australia

SUMMARY

An index is described expressing the similarity between two individuals characterized by a large number of attributes. For each attribute, an ordering relationship between pairs of values is established according with everyday concepts of similarity, but also taking into account the probability that a random sample of two would have the values in question. The cumulative probability of the observed pair or any more similar pair is then calculated, and the probabilities for different attributes are combined. The similarity index for the pair of individuals is then the complement of this combined probability.

It is shown that, in random data, this similarity index is uniformly distributed on the interval (0, 1). It is computed for some data from plant sociology for comparison with two well-known similarity indices.

RÉSUMÉ

Présentation d'un indice mesurant la ressemblance entre deux individus caractérisés par un grand nombre de critères. Pour chacun des critères, une hiérarchie entre paires de valeurs est établie selon un concept courant de ressemblance, mais en tenant compte toutefois de la probabilité que le hasard puisse donner les valeurs en question.

La probabilité d'obtenir la paire observée ou toute autre paire encore plus similaire est alors calculée. Les probabilités relatives aux différents attributs sont ensuite combinées. L'indice de ressemblance des deux individus du couple est alors la probabilité combinée complémentaire.

Il est montré que pour les données aléatoires, cet indice de ressemblance est distribué uniformément sur l'intervalle 0, 1. L'indice est calculé pour quelques données relatives au domaine sociologique et comparé à deux autres indices de ressemblance bien connus.

1. INTRODUCTION

In the field of numerical taxonomy, and wherever individuals characterized by a number of attributes are compared, a means is often required for combining the evidence provided by these attributes as to the degree of resemblance or dissimilarity between a particular pair of individuals. Various similarity indices for this purpose have

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been proposed, from the time of Jaccard [1901] onwards; they have been discussed at some length by Sokal and Sneath [1963] in a recent monograph.

Most of these similarity indices are appropriate only for attributes in binary form, though attempts have been made to modify them so that multi-value attributes can be taken into account. These modifications have usually involved coding the different values for a single attribute as a number of binary attributes (Sneath [1957]; Sokal & Sneath [1963]). The various methods of coding used have the disadvantage either of giving different weight to the attributes according to the number of different values they may take, or of losing information derivable from the ordering of the different values.

Other similarity indices have, however, been proposed to take account of quantitative attributes. For instance, the mean character difference of Cain and Harrison [1958], and the coefficients used by Kendrick and Proctor [1964] and by Sheals [1965], involve calculation for each attribute of the ratio of the difference in values in the two individuals to the maximum value or to the extreme range in the whole population considered, and then averaging these ratios over all attributes. If, however, quantitative and binary attributes are included in the same index, these procedures will generally give the latter excessive weight. These indices appropriate to quantitative characters have from time to time also been applied to ordered, non-quantitative attributes; this can only be done, however, by arbitrarily assigning metric values to their different ordered states.

Similarity Indices and Probability

In most cases where similarity indices have been used, there is a clear stochastic element in the attributes recorded for the two 'individuals' compared. This is perhaps clearest where the 'individuals' are assemblages of organisms and each 'attribute' is the presence of a particular species. The number of representatives of each species in the sample can usually be represented by a Poisson or an overdispersed distribution, so that the chance of complete absence of the species can be ignored only in the most abundant species. Where the similarity of individual organisms is being assessed on the basis of morphological or physiological characters, these too are subject to random processes, being dependent on the organism's genotype with the random effects of mutation and reassortment of genes. Where entire taxa are being compared, the stochastic element is less self-evident, for the attributes used are those common to all or most individuals in the taxon, and thus dependent in part on the activities of

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taxonomists. It might perhaps be argued that natural selection would eliminate the stochastic element from the configuration of attributes separating the taxa—that all would be in greater or lesser measure adaptive. But this is not by any means universally true. The well-recognised Sewall Wright effect ensures that if the effective breeding population has at any stage in the development of the taxa been small, there will be an accumulation of random genetic differences between them. Apomixis and other mechanisms restricting interbreeding likewise make it possible for an assortment of attributes largely random in nature to characterize a taxon.

Even outside the biological field, the subject matter of numerical taxonomy usually includes stochastic features. Potsherds and other artifacts studied by archaeologists are not identical even when from the same source, and many of the attributes which characterize them vary in their expression from example to example. The same is true in such fields as literary criticism. Indeed it is difficult to suggest any subject matter for numerical taxonomy in which none of the attributes recorded could be regarded as in part the result of random processes. If the 'individuals' were a sample from a population in which the various attributes occurred randomly and independently, each with its own frequency, any index of similarity between two 'individuals' would clearly be a stochastic variate, and it seems not unreasonable to choose an index which will reflect the probability in terms of this model.

The most commonly used similarity indices—that of Jaccard, and the 'simple matching coefficient' of Sokal and Michener [1958]—do this very poorly. These coefficients are:

$$\text{Jaccard coefficient} = a/(a + b + c)$$

$$\text{Simple matching coefficient} = (a + d)/(a + b + c + d),$$

where a and d are the numbers of attributes possessed by both individuals and neither individual respectively, and b and c are the numbers of attributes possessed by only one or the other individual. Now consider a population characterized by eight attributes of which numbers 1 to 4 occur in half the population, 5 and 6 in 5% only, and 7 and 8 in 95%. Two individuals of the population have three of these attributes each, of which two are common to both. The Jaccard index is consequently 0.5, and the simple matching coefficient is 0.75. But this situation can arise in various ways. If the two attributes common to both are 7 and 8, while the two which are distinct are 5 and 6, the probability on the hypothesis above is 0.98. If on the contrary the common attributes are 5 and 6, and the distinct ones 7 and

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8, the probability is 0.01. A third such set of attributes—1 and 2 in common, 7 in one and 8 in the other—gives a probability of 0.99.

A lower value of the similarity index may often correspond with a less probable degree of similarity and vice versa. If, for instance, 5 and 6 of the example above were present in both the individuals, while one had 1, 2 and 7, the other 3, 4 and 8 (i.e. $a = 2$, $b = c = 3$), both similarity indices would take values of only 0.25, yet the probability of this degree of similarity between the two would be 0.05, indicating a more significant similarity in this case than in all but one of those considered in the preceding paragraph.

In a recent study on the classification of leukaemias (Hayhoe *et al.* [1964]), the Jaccard similarity index was modified to take account of the differing frequency of the various attributes. The possible frequency values were divided into nine equal ranges. For attributes in the first range (i.e. occurring in less than one-ninth of the cases), positive agreement was given a weight of nine, negative agreement (attribute absent in both cases) was given a weight of one; for attributes in the second range, the weights were respectively eight and two. Disagreement was always given a weight of ten. Thus agreement in respect of half the attributes recorded could give values of the index ranging from 1/11 to 9/19, according to the frequencies of the different states of the attributes in respect of which they agreed.

Distance measures (Sokal [1961]; Sokal and Sneath [1963]) may lend themselves to probabilistic treatment if the distribution of the individual attributes is known. If the individual values have been standardized, larger values of the distance measure will correspond with a less probable degree of similarity. But the actual distribution of the distance measures on the null hypothesis of independence of the attributes will in general not be known.

Ghent [1963] has used Kendall's τ coefficient as an index of similarity between plant and animal communities, the species present being ranked in respect of their quantity in some common metric. This would, of course, lend itself to probabilistic treatment, but could not be applied where the various attributes are incommensurable.

2. THE PROPOSED INDEX

Similarity in respect of single attributes

For our purpose, similarity will be considered as a relation between pairs of individuals within a specified population, although generalization to groups is clearly possible; it can also be applied to the pair of values for a single attribute in two individuals. For any

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particular attribute, similarity will be defined in such a way that it generates an ordering relationship between pairs of values so that, for two pairs of values (a, b) and (c, d), either (a, b) are more similar than (c, d), or the reverse is true, or the two pairs are equally similar. The definition used will depend on the type of attribute in question—whether metrical or non-metrical, ordered or unordered. It will be consistent with ordinary usage, but where the latter provides no guidance in ordering two pairs their order will be based on probability, the more uncommon type of agreement being taken to indicate greater similarity. This rule accords with the intuition of the taxonomist, who regards agreement in possessing an uncommon value for an attribute as more convincing evidence of affinity than agreement in having a common value.

Once all pairs of values for an attribute have been ordered in respect of similarity, the similarity index for each pair is defined as the complement of the probability that a random sample of two will have a similarity equal to, or greater than, the pair in question.

(a) *Qualitative attributes.* The first type of attributes to be considered is that which is purely qualitative, the different values not being capable of being ordered. Here, pairs of differing values are all regarded as equally dissimilar, but pairs of values which agree are ordered according to the rule given above—that is, agreement between two individuals in possessing an uncommon value of the attribute is considered as indicating closer similarity than agreement in possessing a commoner value. The possible pairs of values having been ordered in this way, their probabilities are then calculated, and the measure of similarity for any pair is then the complement of the sum of the probabilities for all pairs of values equal or greater in similarity. In the Appendix this procedure is spelt out formally and in detail, and illustrated by a hypothetical example.

Binary attributes are, of course, special cases of qualitative attributes, and are treated by the same procedure.

(b) *Ordered attributes.* Where the values which an attribute can take form an ordered series, this order should be taken into account in assessing the similarity of individuals in respect of this attribute. As pointed out above, this is not done satisfactorily by the methods of coding commonly used to convert such values into binary form for inclusion in a Jaccard similarity index or matching coefficient. For the moment, only non-metrical attributes will be considered (or attributes in which it is proposed to ignore the actual metrical value recorded, and to treat the records simply as a ranked series.)

Ordinary usage requires that a pair of values included between

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another pair should, *ipso facto*, be regarded as more similar—that is, if the ordered series of values is represented by the letters of the alphabet, (*B*, *C*) are more similar than (*A*, *C*) or (*B*, *E*), and both the latter are more similar than (*A*, *E*). But this does not lead to a complete ordering of pairs: (*A*, *C*) and (*B*, *E*), for instance, remain unordered. For such comparisons we again take probability into account. The similarities of two pairs neither of which includes the other are ordered by the sum of probabilities of all possible pairs of values which each includes—or, to put it another way, by the proportion of the population which each encompasses. Thus, if the proportions of individuals in four ordered categories were 0.1, 0.4, 0.4 and 0.1, two pairs (1, 3) and (2, 4) would be regarded as equally similar because each encompasses 0.9 of the population. On the other hand, a pair (1, 2) would be regarded as more similar than a pair (2, 3) since they encompass respectively 0.5 and 0.8 of the population.

Where an ordered attribute falls into discrete classes, two individuals with the same value, whatever this may be, will usually be regarded as more similar than two with differing values. More often, however, it may be better regarded as varying continuously, even though the exigencies of observation may lead to the values being grouped in a small number of classes. In this case, it may not be justifiable to treat two individuals with the same value as necessarily more similar than two with differing values—there may be differences in discrimination in different parts of the underlying scale. In this case, all pairs of values, equal or unequal, are treated alike and are ordered according to the proportion of the population encompassed by the two values.

Once all pairs of values have been ordered in this way, determination of similarity follows the same pattern as for qualitative attributes. The whole procedure is set out formally in the Appendix.

(c) *Metrical Attributes.* Where the values taken by an attribute are not merely ordered but metrical, the appropriate criterion for ordering pairs in respect of similarity is simply the difference between the two values. Pairs with identical values are more similar than those which differ, those with a small difference are more similar than those with a larger difference. As between pairs of values differing by the same amount, however, it appears desirable to take into account the size of the groups encompassed by the two values, in much the same way as was done with ordered attributes.

Again, once these ordering relations have been established, the degree of similarity is expressed by the complement of the probability of the observed pair or any more similar. The derivation is given in

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detail in the Appendix for an attribute taking a number of discrete metrical values.

Where the value of an attribute is a continuous variable whose distribution is known, it may be practicable to make use of the distribution of the range in samples of two. Otherwise, one can proceed empirically by testing the range of the pair in comparison with each pair of individuals in the whole set of data; but except for small numbers of individuals and attributes this would be impracticable. As an approximation, one may divide the distribution into a convenient number of groups, and use the mean or median of each group as the value of the attribute for each individual included in it.

In some fields, a case can be made for the view expressed by Williams and Dale [1962] that a zero value for a metrical attribute is to be treated rather as a distinct qualitative character than simply as a limiting value. If this view is taken, such an attribute might well be treated as two—the one a two-value qualitative attribute indicating presence or absence, the other a metrical attribute becoming indeterminate when the first is zero. Even in such a case, though, it seems not unreasonable to regard an individual with a small positive value for the attribute as somewhat more akin to those in which it is zero than are others in which the value is large. In this event, it may be better to add to all positive values an amount equal to the largest recorded, and then treat as a straightforward metrical attribute.

Combination of similarities in respect of different attributes

Up to this point, each attribute has been considered separately, and it is now necessary to discuss the combination of data in respect of a number of attributes. The assumption is made that the values taken by different attributes in the same individual are independent. If the population is heterogeneous, this will not be true; the heterogeneity will then be reflected in the distribution of the similarity indices, which thus provide a test of homogeneity.

In order to combine similarities in respect of different attributes, it is again necessary to define an ordering relation between pairs—not now between pairs of values of a single attribute, but between pairs of sets of values for the various attributes to be included. Again, we have recourse to probability. If, for each attribute, all pairs of values have been ordered in respect of similarity and cumulative probabilities have been calculated for each pair, the product of these cumulative probabilities over all attributes provides an ordering relation between pairs of sets of values. Those pairs for which this product is smaller

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are more similar, and *vice versa*. Then, as with separate attributes, the individual probabilities are summed over all pairs of sets not less similar than that under consideration to give a measure of dissimilarity, of which the similarity index is the complement. It is defined formally in the Appendix, and its calculation is illustrated by a simple example.

On examining this example it will immediately be apparent that the computational load imposed by this exact method of combining probabilities would mount very rapidly with increase in the number of attributes t and alternatives n_i $(1 \leq i \leq t)$, soon becoming impracticable even for an electronic computer. Fisher's solution [1963] for the combination of continuous probabilities from independent tests may be used as an approximation instead. Where a number t of independent probability estimates P_i are available, a quantity is calculated

$$-2 \sum_{i=1}^{t} \ln P_i$$

which is distributed as χ^2 with $2t$ degrees of freedom. The probability of this χ^2 statistic, or any greater value, may be obtained from tables or computed from the expression

$$P = e^{-\chi^2/2} \sum_{i=0}^{t-1} (\tfrac{1}{2}\chi^2)^i / i!$$

to give a value P which is the overall probability of obtaining a set of t independent probabilities collectively as improbable as that observed.

If applied to discrete probabilities this method gives a P with a positive bias which may be very large; an improvement was proposed by Lancaster [1949] in which each P_i is replaced by a value intermediate between the probability of the event actually observed and the next smaller probability in the discrete set including it. If this smaller probability is designated P_i', Lancaster's modified χ^2 statistic may be written

$$2 \sum_{i=1}^{t} \left\{ 1 - \frac{P_i \ln P_i - P_i' \ln P_i'}{P_i - P_i'} \right\}.$$

The probability of this modified χ^2 statistic appears to be free from bias except at the extremes, though it may deviate considerably from the true combined probability. To illustrate these deviations, let us consider two probabilities each from a set of three possible values, as follows:—

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$$p_1 = .01 \qquad p_1 = .001$$
$$p_2 = .39 \qquad p_2 = .039$$
$$p_3 = .60 \qquad p_3 = .960.$$

The event (1, 2) and the event (2, 1) give exactly the same value, 0.00079, for the exact probability of the observed or any less likely event, but the Lancaster χ^2 statistics are 17.6 and 19.0 respectively, yielding probabilities of 0.0015 and 0.0008.

In applying to similarity indices this method of combining probabilities, P_i' is clearly to be taken as the next smaller probability of the discrete observable set $P_{k_i l_i}$ (see Appendix) for each attribute. In order to reduce the deviations of the estimated from the true combined probabilities, one may combine exactly the $P_{k_i l_i}$ for a small number of attributes before using Lancaster's χ^2 approximation, and thus compromise between the usually impracticable time requirement for the exact solution, and the large deviations which may result when the Lancaster approximation is applied to probabilities from very small discrete sets.

As an illustration of the effect of grouping, we may instance a fictitious set of data (Table 1) for six two-value attributes in twelve individuals. The combined probabilities $\mathcal{P}_{kl}$ have been calculated exactly, and the separate probabilities $P_{k_i l_i}$ have also been combined by the Lancaster approximation either singly, or after grouping in twos or threes. Some of the results are shown in Table 2.

Since the estimate given by the Lancaster approximation depends on the particular discrete sets of probabilities combined, and is by no means constant for any particular value of $\mathcal{P}_{gh}$ made up in different

TABLE 1
A Hypothetical Record of Two-value Attributes

	Individual											
Attribute	1	2	3	4	5	6	7	8	9	10	11	12
A	1	0	0	0	0	1	0	0	0	0	0	0
B	1	0	0	1	0	1	0	1	0	0	0	0
C	1	1	1	1	0	1	1	1	1	1	0	0
D	0	0	1	1	0	1	0	0	0	1	0	1
E	0	0	0	1	1	0	1	0	0	0	0	0
F	1	0	1	0	0	1	0	1	1	0	1	1

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TABLE 2
SOME INTER-INDIVIDUAL COMPARISONS FROM TABLE 1

Pairs of individuals		$\mathcal{P}_{gh}$	$\mathcal{P}_{gh}$ estimated by Lancaster approximation				
			$P_{g_ih_i}$ singly	$P_{g_ih_i}$ grouped in twos		$P_{g_ih_i}$ grouped in threes	
				(1)	(2)	(1)	(2)
1	11	.66283	.71125	.61487	.58436	.61815	.65749
2	7	.16539	.17157	.16288	.23481	.13715	.14977
4	5	.06803	.10859	.04369	.08730	.04512	.07896
5	7	.01792	.03746	.01104	.03991	.00977	.02358
6	7	.97752	.94496	.97143	.97016	.95647	.97446
7	9	.55105	.50044	.45285	.67805	.42533	.54541
9	11	.35763	.36127	.30206	.30309	.37062	.29525
10	12	.29640	.25967	.27640	.20193	.29926	.26524

ways, when the attributes are grouped before applying the Lancaster method the results will depend on which particular estimates are combined in groups. This is illustrated in Table 2. The columns headed (1) were calculated for the attributes arranged as shown in Table 1; for those headed (2), the order of the attributes was changed so that different sets (twos or threes) were combined before the Lancaster approximation was applied. It will be noted that the order of grouping has affected the estimated $\mathcal{P}_{gh}$; for the smaller probabilities, the proportional difference is sometimes considerable. Grouping in threes has usually reduced the magnitude of the difference due to the way in which the attributes were combined, as compared with grouping in pairs. In general, where more attributes are available and $P_{g_ih_i}$ can take a larger set of values, estimates are likely to be affected much less by the way in which the attributes are grouped.

The desirability of grouping clearly depends on the number of alternative values the similarity between two individuals for a given attribute can take. For two-value attributes the number of alternative similarity values is (except in degenerate cases) three. As an arbitrary rule, it is proposed that the number of alternative values available to any probabilities it is proposed to combine by the χ^2 method should not be less than ten. This means that probabilities for two-value attributes should be combined in groups of three or more, and those for multi-value attributes in groups of two except that ordered or quantitative attributes taking more than four different values do not in general require grouping.

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3. EXAMPLES

A test on random data

A test of the proposed similarity index has been performed on a model population of twenty individuals characterized by twenty attributes, as follows:—

Attributes 1– 4: Two-value attributes.
Attributes 5– 8: Qualitative attributes, taking 4, 4, 8 and 9 different values respectively.
Attributes 9–12: Ordered attributes taking 4, 4, 9 and 10 different values respectively; these were taken not as discrete classes, but as arbitrary segregates in a continuously ordered underlying series (see §2(b), p. 886).
Attributes 13–16: Quantitative attributes taking 4, 4, 9 and 10 different values respectively.
Attributes 17–20: Quantitative attributes taking different values in each individual.

Attribute values were allotted to the individuals by reading from a table of random numbers.

Similarity indices between all pairs of individuals of this model population were calculated, either combining the probabilities for the separate attributes directly by the χ^2 method, or grouping in twos or threes to determine exact probabilities beforehand. Figure 1 shows the distribution of the 190 similarity indices in each set. The true probabilities, for a random model population, would of course form a rectangular distribution on the interval (0, 1). These data answer this test satisfactorily. Even the ungrouped indices, which seem to have a slight excess of values around 0.6 and a deficiency of very high and very low values, do not deviate significantly from the rectangular distribution.

An example from plant ecology

This proposed similarity index will now be applied to some data for eight grassland communities in Colorado which Hanson [1955] compared by use of the Sørensen [1948] similarity coefficient. A total of 89 species were present in the eight communities—from 23 to 49 in each separately—and estimates of the percentage of ground covered by each species were obtained. Hanson used these cover estimates to compute similarity coefficients by the Sørensen method, and his results, which he expressed in percentage terms, are reproduced in Table 3. For comparison, his presence and absence data have also been used to compute Jaccard and Sørensen indices, similarly expressed as

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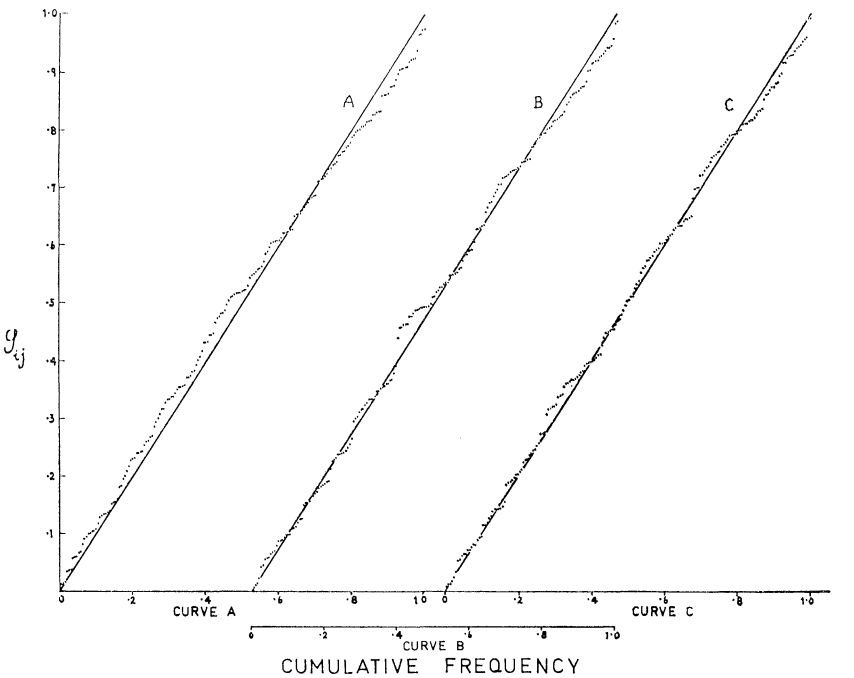


FIGURE 1
CUMULATIVE DISTRIBUTION OF SIMILARITY INDICES FOR RANDOM DATA. ATTRIBUTES UNGROUPED (CURVE A), GROUPED IN TWOS (CURVE B), OR GROUPED IN THREES (CURVE C).

percentages in Tables 4 and 5. The same data have been used to compute the new similarity index, in various ways. For Table 6, only the presence or absence of the species in any community is taken into account as a two-value attribute. For Table 7 the cover values were

TABLE 3
SØRENSEN SIMILARITY INDICES, BASED ON COVER, REARRANGED FROM HANSON'S [1955] TABLE III

Stand	F_1	F_2	F_4	F_5	F_6	F_7	F_8
F_2	66.1						
F_4	68.7	76.0					
F_5	50.2	54.6	61.0				
F_6	49.7	64.3	62.6	62.6			
F_7	45.7	44.3	53.0	66.8	64.2		
F_8	61.9	71.9	71.9	57.2	62.0	52.4	
F_9	61.7	58.6	62.1	54.7	53.0	53.0	69.4

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divided into the seven categories 0, x (present, but too sparse for cover to be measured), 0.1 − 0.2, 0.3 − 0.5, 0.6 − 1.0, 1.1 − 2.0, >2.0 and treated as semi-quantitative (ordered) attributes. For Table 8, the difference between cover values was compared with every other difference between two individuals to give the similarity index, x here being counted as 0.01 per cent.

TABLE 4

JACCARD SIMILARITY INDICES FOR HANSON'S DATA, BASED ON PRESENCE

Stand	F_1	F_2	F_4	F_5	F_6	F_7	F_8
F_2	40.5						
F_4	37.0	46.2					
F_5	28.5	46.6	50.8				
F_6	32.0	46.3	45.6	50.8			
F_7	30.6	37.5	44.6	52.5	52.7		
F_8	30.6	45.3	39.7	40.6	44.8	39.0	
F_9	37.0	35.7	40.4	48.3	40.7	44.6	47.3

TABLE 5

SØRENSEN SIMILARITY INDICES FOR HANSON'S DATA BASED ON PRESENCE

Stand	F_1	F_2	F_4	F_5	F_6	F_7	F_8
F_2	57.6						
F_4	54.0	63.2					
F_5	44.4	63.5	67.4				
F_6	48.5	60.8	62.7	67.4			
F_7	46.9	54.5	61.7	68.9	69.0		
F_8	46.9	62.3	56.8	57.8	61.9	56.1	
F_9	54.0	52.6	57.5	65.2	57.8	61.7	64.2

TABLE 6

PROBABILISTIC SIMILARITY INDICES FOR HANSON'S DATA BASED ON PRESENCE

Stand	F_1	F_2	F_4	F_5	F_6	F_7	F_8
F_2	.944						
F_4	.839	.713					
F_5	.009	.795	.908				
F_6	.097	.669	.253	.513			
F_7	.168	.115	.555	.884	.894		
F_8	.168	.592	.126	.237	.244	.197	
F_9	.724	.091	.660	.800	.026	.634	.906

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TABLE 7
PROBABILISTIC SIMILARITY INDICES FOR HANSON'S DATA BASED ON COVER IN ARBITRARY ORDERED CATEGORIES

Stand	F_1	F_2	F_4	F_5	F_6	F_7	F_8
F_2	.771						
F_4	.641	.761					
F_5	.032	.653	.653				
F_6	.121	.751	.503	.958			
F_7	.417	.009	.724	.984	.994		
F_8	.274	.547	.158	.103	.045	.184	
F_9	.673	.246	.502	.608	.024	.316	.969

It is clear that, whereas the Jaccard and Sørensen indices give values concentrated near the middle of the range (the extreme values in Tables 3–5 are respectively 45.7–76.0; 28.0–50.8; and 44.0–68.9), the new index extends over the whole range from 0 to 1, and in fact has somewhat more values towards these limits than in the central part of the range. This is to be expected where there are real differences between the 'individuals' (stands) compared, so that the assortment of 'attributes' (species) among them has not in fact been random but dependent on common underlying features (environmental factors) affecting a number of them.

Despite the difference in distribution of the indices tabuated, all of them show fairly close association. The product-moment correlation coefficient r and Kendall's rank correlation τ between the 28 indices have been computed for each pair of these six tables; the results are shown in Table 9. Since the Jaccard and Sørensen presence indices make use of the same values of a, b, and c, they are extremely highly correlated. But the new index (Table 6) based on presence is

TABLE 8
PROBABILISTIC SIMILARITY INDICES FOR HANSON'S DATA BASED ON COVER AS A QUANTITATIVE VARIABLE

Stand	F_1	F_2	F_4	F_5	F_6	F_7	F_8
F_2	.914						
F_4	.792	.925					
F_5	.369	.348	.774				
F_6	.358	.942	.099	.467			
F_7	.133	.001	.094	.395	.969		
F_8	.515	.967	.816	.091	.053	.043	
F_9	.943	.643	.545	.530	.122	.197	.988

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TABLE 9
CORRELATIONS AMONG SIMILARITY INDICES FOR HANSON'S DATA CALCULATED BY DIFFERENT METHODS
(r below the principal diagonal, τ above it)

Indices from Table number	4	5	6	7	8	9
4	1.0000	.3120	.3120	.3757	.4127	.5079
5	.4360	1.0000	1.0000	.5093	.4613	.1947
6	.4409	.9988	1.0000	.5093	.4613	.1947
7	.5031	.6674	.6640	1.0000	.6138	.3810
8	.5298	.6644	.6516	.8583	1.0000	.4286
9	.7392	.2312	.2253	.5977	.5846	1.0000

also quite highly correlated with these two, and the new index based on cover as a quantitative variable (Table 8) is also highly correlated with the Sørensen index based on the same data (Table 3). Correlations between indices based on cover and those based on presence are often fairly low; the index based on categorised cover data (Table 7) takes an intermediate position, its correlation being reasonably high both with the presence indices, and with those of Tables 3 and 8.

Most of Hanson's conclusions based on the Sørensen similarity indices of Table 3 are confirmed by study of those in Table 8. For instance he divides the quadrats into two groups—F_1 , F_2 , F_4 , F_8 and F_9 ; and F_5 , F_6 and F_7 . In Table 8, all the indices for the first group exceed .515. Since there are 56 possible subsets of 5 in a total of 8, the probability of such a grouping, on the assumption that all stands were samples of the same community with random assortment of species among them is about .05.[1] The second group (F_5 , F_6 and F_7) do not appear to be as closely related; between the two groups however, most of the indices are low—all but one of the fifteen are less than .55.

4. DISCUSSION

It is clear that this index forms a useful addition to those currently in use for numerical taxonomy. It can be used in entirely comparable fashion for all classes of attribute, and is not dependent, as are some of the indices currently in use, on binary coding, which almost inevitably results in a loss of information. It further enables the rarer characteristics to be given the enhanced weight which the working taxonomist

[1]This figure is based on the assumption of independence among the similarity indices; though not strictly true, this is often a reasonable approximation, as will be shown in a forthcoming publication.

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prefers. Moreover, it clears the way to a complete procedure for numerical taxonomy controlled by specified significance levels. Procedures for this purpose are now being developed (see, for instance, Goodall [1966]).

An anomaly in some of the commonly used similarity indices arises from the treatment of attributes present in the whole set of individuals under consideration. With many types of subject-matter such attributes can be multiplied *ad libitum*, and their inclusion can increase the similarity index to any desired extent. This introduces into the absolute value of the index an undesirable arbitrary element, which some authors have avoided by excluding all such attributes from consideration. When this is done, the similarity of any pair of individuals is clearly not absolute, but dependent on the set of individuals within which the pair is considered. The present index, unaffected as it is by attributes taking the same value in all individuals, likewise depends on the context in which the pair of individuals is considered. Unless external information as to the population frequencies of attribute values is available, they must be estimated from the whole set of individuals under consideration, regarded as a sample of the population. Consequently, any change in the set of individuals will affect all the similarities within it. As Williams and Lance [1965] remark, this reflects a very common taxonomic outlook; 'similarity' is to be regarded as having a meaning only within a particular context, and not absolutely.

Another anomaly arises where, as in the Jaccard and Sørensen indices, presence and absence of attributes are treated differently. It is often a purely arbitrary choice which of the values of a two-value attribute should be treated as 'presence'—the presence of an enzyme, for instance, may be the phenotypic expression of absence of a repressor gene (cf. Hill *et al.* [1961]); yet the value of the similarity index may be greatly affected by such choices. In the index proposed here the question does not arise, for presence and absence are simply treated as two alternative values of a qualitative attribute.

A difficulty recognised by many writers on numerical taxonomy is the treatment of logically dependent attributes. Where, for instance, one attribute considered is the presence of an organ, and another is its shape, the second is clearly indeterminate in the absence of the first. In the present method, it is often possible to include both as different values of a single attribute. For instance, if the petals of a flower were either present or absent in different species, and if present they might be either notched or entire, the observations could be presented as three alternative values of a single attribute. On the other

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hand, it might with some justice be argued that two plants with petals respectively notched and entire were more like one another than either was like another plant completely devoid of petals—a point of view not satisfactorily covered by treating them as three values of a single attribute, whether qualitative or ordered. It might perhaps be thought that the dilemma could be resolved by including presence and shape as two separate attributes; if this is done, however, the process of combining probabilities will take into account combinations of values (for instance, both absent, and both notched) which are *ex hypothesi* impossible, and consequently the combined probabilities (whether calculated exactly, or by the approximation) will be biased. In such cases, a special combination procedure seems called for, taking into consideration only those sets of attribute values which are permitted by their inter-dependence.

Empirical dependence between two attributes—due to a biological rather than a logical relation between them, as for instance the presence of pigmentation in two different organs—cannot be satisfactorily handled by this, or indeed any other similarity index. It is an assumption of the method that the attributes are independent, and a biological relationship of this sort will show itself as an enhanced similarity between those possessing, or not possessing, these attributes.

The computational load imposed by this index is considerable—in fact, its use could hardly be considered, except in the simplest cases, without an electronic computer. The computation time clearly depends on the types of attributes included and the number of alternatives for each of them, as well as on the number of attributes and individuals. It also depends very markedly on the size of the attribute groups which are combined exactly to give the probabilities combined by the χ^2 procedure. If this group size is a, the number of attributes b, and the number of individuals c, the computation time required for a complete matrix of similarity indices is given very roughly by

$$3^a bc^2 k$$

where k is a constant depending on computer and programme. Calculations have so far been performed with a rather inefficient programme in FORTRAN II (which can be made available to enquirers) on three machines—an IBM 1620 with an integer addition time of 480 μsec., a PDP6, with an integer addition time averaging about 4 μsec. and a CDC 3600, for which the integer addition time is 2.07 μsec. The values of k are approximately 60 msec., 250 μsec. and 200 μsec. respectively.

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APPENDIX

The calculation of the similarity index is here described in detail, under the same sub-headings as in the text. Similarity is defined for each of the three types of attribute in question, and the combination of these separate similarities is then considered. To simplify the presentation, notation from symbolic logic and set theory is used on occasion, including the following symbols:—

$\supset$: implies

$\wedge$: and

$\vee$: inclusive or

$\in$: is included in

(a) *Qualitative attributes*

Let the alternative values V_i of a purely qualitative attribute occur in the proportions p_i , $1 \leq i \leq n$, $\sum_{i=1}^{n} p_i = 1$. Now if S_{ij} is the similarity between two values for this attribute, ordinary usage requires that

$$i \neq j \supset S_{ij} = 0$$
$$i = j \supset S_{ij} > 0.$$

As described in the text, we add

$$(i = j) \wedge (k = l) \wedge (p_i < p_k) \supset S_{ij} > S_{kl} .$$

If now P_{ij} is the probability that a random pair will have a similarity equal to, or greater than, S_{ij} , we have

$$P_{ij} = \sum_{k \in Q} p_k^2 , \qquad i = j,$$

where

$$Q = \{k: (p_k \leq p_i)\}$$

i.e. the set of all k ($1 \leq k \leq n$ being understood) such that $p_k \leq p_i$; and

$$P_{ij} = 1, \qquad i \neq j.$$

Hence, by definition, the similarity index

$$S_{ij} = 1 - P_{ij} .$$

In the practical situation, of course, the p_i will not be known but need to be estimated from the sample of m individuals. If f_i is the number of individuals in m which have V_i as the value of the attri-

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bute in question, the appropriate estimate of S_{ij} is

$$\hat{S}_{ij} = 1 - \sum_{k \in Q} \frac{f_k(f_k - 1)}{m(m-1)}, \qquad i = j,$$

and

$$\hat{S}_{ij} = 0, \qquad i \neq j,$$

where Q is as defined above, but in terms of the f's rather than the p's.

As an example of this calculation, let us take an attribute with $n = 4$ values, viz.

$$V_1 = \text{red}; \qquad V_2 = \text{yellow}; \qquad V_3 = \text{green}; \qquad V_4 = \text{blue}$$

recorded in a sample of $m = 15$ individuals with frequencies of

$$f_1 = 2; \qquad f_2 = 4; \qquad f_3 = 4; \qquad f_4 = 5,$$

and let us consider a pair of individuals both of which are yellow—for which, in other words, the values for comparison are (V_2, V_2). In this case

$$Q = \{1, 2, 3\}$$

and

$$\hat{S}_{ij} = 1 - \frac{f_1(f_1 - 1) + f_2(f_2 - 1) + f_3(f_3 - 1)}{m(m-1)}$$

$$= 1 - \frac{2 + 12 + 12}{210} = \frac{184}{210}.$$

(b) *Ordered attributes*

Let an attribute take an ordered series of values

$$V_1, V_2, \cdots, V_n.$$

To be consonant with the qualitative attributes considered in the previous section, we require that

$$(i = j) \wedge (k = l) \wedge (p_i < p_k) \supset S_{ij} > S_{kl}$$

and

$$(i = j) \wedge (k \neq l) \supset S_{ij} > S_{kl}.$$

Furthermore, ordinary usage indicates that

$$i < j < k \supset (S_{ij} > S_{ik}) \wedge (S_{jk} > S_{ik}).$$

As described in the text, the ordering of pairs is completed by the criteria

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$$(i \neq j) \wedge (k \neq l) \wedge \left(\sum_{t=i}^{j} p_t = \sum_{t=k}^{l} p_t\right) \supset S_{ij} = S_{kl}$$

and

$$(i \neq j) \wedge (k \neq l) \wedge \left(\sum_{t=i}^{j} p_t < \sum_{t=k}^{l} p_t\right) \supset S_{ij} > S_{kl} .$$

Where all the above criteria are applicable, the probability P_{ij} that a random pair of values will be as similar as, or more similar than, the pair (V_i , V_j) is defined by

$$P_{ij} = \sum_{k \in Q} p_k^2 , \qquad i = j,$$

where

$$Q = \{k: (p_k \leq p_i)\};$$

and

$$P_{ij} = \sum_{k=1}^{n} p_k^2 + 2 \sum_{k=1}^{n-1} \sum_{l=k+1}^{r} p_k p_l , \qquad i \neq j,$$

where r is defined by

$$\sum_{t=k}^{r} p_t \leq \sum_{t=i}^{j} p_t < \sum_{t=k}^{r+1} p_t .$$

Estimating the p_i from observed frequencies f_i , these become

$$\hat{P}_{ij} = \sum_{k \in Q} \frac{f_i(f_i - 1)}{m(m-1)} , \qquad i = j,$$

and

$$\hat{P}_{ij} = \sum_{k=1}^{n} \frac{f_i(f_i - 1)}{m(m-1)} + 2 \sum_{k=i}^{n-1} \sum_{l=k+1}^{r} \frac{f_k f_l}{m(m-1)} , \qquad i \neq j,$$

where Q and r are defined as above, but in terms of the f's rather than the p's.

Where the scale underlying the ordered attribute is regarded as continuous (see text), the criterion

$$(i = j) \wedge (k \neq l) \supset S_{ij} > S_{kl}$$

is abandoned, and the two criteria

$$\sum_{t=i}^{j} p_t = \sum_{t=k}^{l} p_t \supset S_{ij} = S_{kl}$$

and

$$\sum_{t=i}^{j} p_t < \sum_{t=k}^{l} p_t \supset S_{ij} > S_{kl}$$

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suffice for a suitable ordering of all pairs of values. The definitions required are now

$$P_{ik} = \sum_{i \in S} \left\{ p_i^2 + 2 \sum_{i'=i+1}^{r} p_i p_{i'} \right\}$$

where

$$S = \left\{ i : p_i \leq \sum_{t=j}^{k} p_t \right\}$$

and r is defined by

$$\sum_{t=i}^{r} p_t \leq \sum_{t=j}^{k} p_t < \sum_{t=1}^{r+1} p_t \ .$$

Where the p_i have to be estimated from observed frequencies f_i, the expression becomes

$$\hat{P}_{ik} = \sum_{i \in S}^{n} \left\{ \frac{f_i(f_i - 1)}{m(m-1)} + 2 \sum_{l=i+1}^{r} \frac{f_i f_l}{m(m-1)} \right\},$$

where S and r are defined as above, but in terms of the f's.

Taking as an example a case where the attribute is recorded in five categories for 100 individuals, with frequencies $f_1 = 5$, $f_2 = 10$, $f_3 = 30$, $f_4 = 30$, $f_5 = 25$, for the fourth and fifth classes the $\hat{P}_{ik}$ would be the sum of estimated probabilities that a random pair of values should be in the following classes or pairs of classes:—

Class	$\frac{f_i(f_i-1)}{m(m-1)}$	Classes	$\frac{f_i f_l}{m(m-1)}$
1	.0020	1,2	.0101
2	.0091	1,3	.0303
3	.0879	2,3	.0606
4	.0879	4,5	.1515
5	.0606		

That is, $\hat{P}_{45} = .500$.

(c) *Metrical Attributes*

Our ordering relationship is here defined by

$$|V_i - V_j| < |V_k - V_l| \supset S_{ij} > S_{kl}$$

$$(|V_i - V_j| = |V_k - V_l|) \wedge \left(\sum_{t=i}^{j} p_i < \sum_{t=k}^{l} p_i \right) \supset S_{ij} > S_{kl}$$

$$(|V_i - V_j| = |V_k - V_l|) \wedge \left(\sum_{t=i}^{j} p_i = \sum_{t=k}^{l} p_i \right) \supset S_{ij} = S_{kl} \ .$$

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Accordingly, we define

$$P_{jk} = \sum_{i \in Q} \{p_i^2 + 2 \sum_{t \in T_i} p_i p_t\}$$

where

$$Q = \{i: [(j \neq k) \vee (p_i \leq p_j)] \wedge [1 \leq i \leq n]\}$$

and

$$T_i = \left\{t: \left[(|V_t - V_i| < |V_k - V_j|) \vee \left[(|V_t - V_i| = |V_k - V_j|) \wedge \left(\sum_{u=j}^{k} p_u \geq \sum_{u=i}^{t} p_u\right)\right]\right] \wedge [i < t \leq n]\right\}$$

This may be estimated from sample data by

$$\hat{P}_{jk} = \frac{1}{m(m-1)} \sum_{i \in Q} \{f_i(f_i - 1) + 2 \sum_{t \in T_i} f_i f_t\}$$

where Q and T_i are defined as above, but in terms of the f's.

Again taking a simple numerical example in the hope of clarifying this presentation, let us suppose a sample of $m = 15$ individuals, for which the values of the attribute and the corresponding frequencies are:—

$V_1 = -3$	$f_1 = 4$
$V_2 = -1$	$f_2 = 1$
$V_3 = 0$	$f_3 = 2$
$V_4 = 2$	$f_4 = 2$
$V_5 = 5$	$f_5 = 6$

and let us consider a pair of individuals with values of 0 and 2 respectively; in other words $j = 3$, $k = 4$. Then

$Q = \{1, 2, 3, 4, 5\}$

T_1 is empty, for $(V_2 - V_1) = (V_4 - V_3)$ but $\sum_{u=3}^{4} f_u < \sum_{u=1}^{2} f_u$

$T_2 = \{3\}$, for $(V_3 - V_2) < (V_4 - V_3)$

$T_3 = \{4\}$

T_4 is empty, for $(V_5 - V_4) < (V_4 - V_3)$

T_5 is empty

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so

$$\hat{P}_{ik} = \frac{1}{210}\{4.3 + (1.0 + 2.1.2) + (2.1 + 2.2.2) + 2.1 + 6.5\}$$
$$= 58/210.$$

Combination of similarities in respect of different attributes

Let the values of the attributes for the two individuals in question be expressed by the vectors

$$\mathbf{V_g} = (V_{gi}); \qquad \mathbf{V_h} = (V_{hi}), \qquad 1 \leq i \leq t$$

and the proportions of the different values of the ith attribute be

$$p_{ij_i}, \qquad 1 \leq j_i \leq n_i .$$

Again, it is necessary to define a relationship which will order all possible pairs of individuals in respect of their similarity. In conformity with the previous section, we write

$$P_{g_ih_i}$$

for the probability that a random pair of values for the ith attribute will be at least as similar as the values observed for this attribute in individuals g and h. Then we define the overall similarity $\mathbb{S}_{gh}$ between these individuals by using the criterion

$$\prod_{i=1}^{t} P_{g_ih_i} < \prod_{i=1}^{t} P_{k_il_i} \supset \mathbb{S}_{gh} > \mathbb{S}_{kl}$$

and again summing the probabilities of all possible pairs of vectors $(\mathbf{k}, \mathbf{l})$ as similar as, or more similar than, the pair $(\mathbf{g}, \mathbf{h})$ to give

$$\mathcal{P}_{gh} = 1 - \mathbb{S}_{gh} .$$

If, then, possible sets of values for the t attributes in two individuals are defined by the vectors

$$\mathbf{V_k} = (V_{ki}); \qquad \mathbf{V_l} = (V_{li}), \qquad 1 \leq i \leq t$$

the probability that two random selections of values for the attributes from the population specified will not be less similar than $(\mathbf{V_g}, \mathbf{V_h})$ is

$$\mathcal{P}_{gh} = \sum_{k_1=1}^{n_1} \sum_{l_1=1}^{n_1} \sum_{k_2=1}^{n_2} \sum_{l_2=1}^{n_2} \cdots \sum_{k_t=1}^{n_t} \sum_{l_t=1}^{n_t} \prod_{i=1}^{t} \prod_{j=1}^{t} p_{k_i} p_{l_j}$$

where summation extends only over such values of the vectors $\mathbf{V_k}\mathbf{V_l}$ as meet the conditions

$$\prod_{i=1}^{t} P_{k_il_i} \leq \prod_{i=1}^{t} P_{g_ih_i} .$$

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As an example, let us suppose that two qualitative attributes take the values V_{1_1} , V_{2_1} , V_{3_1} and V_{1_2} , V_{2_2} , V_{3_2} with the probabilities

$$p_{1_1} = .1 \qquad p_{1_2} = .2$$
$$p_{2_1} = .2 \qquad p_{2_2} = .3$$
$$p_{3_1} = .7 \qquad p_{3_2} = .5.$$

TABLE 10
COMPUTATION OF A HYPOTHETICAL $\mathcal{P}_{gh}$

V_k , V_l	$\prod_{i=1}^{2}\prod_{j=1}^{2} p_{k_i}p_{l_j}$	$\prod_{i=1}^{2} P_{k_i l_i}$
$(V_1 , V_1); (V_1 , V_1)$	.0004	.0004
$(V_1 , V_1); (V_1 , V_2)$	.0012	.0100
$(V_1 , V_1); (V_1 , V_3)$	.0020	.0100
$(V_1 , V_1); (V_2 , V_1)$	.0016	.0400
$(V_1 , V_1); (V_3 , V_1)$	.0056	.0400
$(V_1 , V_2); (V_1 , V_2)$	.0009	.0013
$(V_1 , V_2); (V_1 , V_3)$	.0030	.0100
$(V_1 , V_2); (V_2 , V_2)$	.0036	.1300
$(V_1 , V_2); (V_3 , V_2)$	.0126	.1300
$(V_1 , V_3); (V_1 , V_3)$	.0025	.0038
$(V_1 , V_3); (V_2 , V_3)$	.0100	.3800
$(V_1 , V_3); (V_3 , V_3)$	.0350	.3800
$(V_2 , V_1); (V_2 , V_1)$	.0016	.0020
$(V_2 , V_1); (V_2 , V_2)$	.0048	.0500
$(V_2 , V_1); (V_2 , V_3)$	.0080	.0500
$(V_2 , V_1); (V_3 , V_1)$	.0112	.0400
$(V_2 , V_2); (V_2 , V_2)$	.0036	.0065
$(V_2 , V_2); (V_2 , V_3)$	.0120	.0500
$(V_2 , V_2); (V_3 , V_2)$	.0252	.1300
$(V_2 , V_3); (V_2 , V_3)$	.0100	.0190
$(V_2 , V_3); (V_3 , V_3)$	.0350	.3800
$(V_3 , V_1); (V_3 , V_1)$	.0196	.0216
$(V_3 , V_1); (V_3 , V_2)$	.0588	.5400
$(V_3 , V_1); (V_3 , V_3)$	.0980	.5400
$(V_3 , V_2); (V_3 , V_2)$	.0441	.0702
$(V_3 , V_2); (V_3 , V_2)$	.2940	.5400
$(V_3 , V_3); (V_3 , V_3)$	.1225	.2052

For all other sets $\prod_{i=1}^{2} P_{k_i l_i} = 1.0000.$

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This would give as $P_{k_i l_i}$ the following:

$$P_{1_1 1_1} = .01 \qquad P_{1_2 1_2} = .04$$

$$P_{2_1 2_1} = .05 \qquad P_{2_2 2_2} = .13$$

$$P_{3_1 3_1} = .54 \qquad P_{3_2 3_2} = .38$$

$$P_{i_1 j_1} = 1.00, \quad i_1 \neq j_1 \qquad P_{i_2 j_2} = 1.00, \quad i_2 \neq j_2 .$$

The probabilities of the various sets of values which might occur in the two individuals are then those given in Table 10.

If the individuals g and h under consideration agree in respect of both attributes, which have the values V_1 and V_3, the probabilty of this or any more similar pair of attribute sets is then the sum of all the

$$\prod_{i=1}^{2} \prod_{j=1}^{2} p_{k_i} p_{l_i}$$

in Table 10 for which

$$\prod_{i=1}^{2} P_{k_i l_i} \leq .0038,$$

viz.

$$\mathcal{P}_{gh} = .0054,$$

and the index of similarity is accordingly

$$\mathcal{S}_{gh} = 0.9946.$$

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